

1.

For R-CNN, it uses selective search to get many many regions before passing them through CNN, and each region is passed independently, so CNN is ran many many times, therefore the feature extraction is very very slow. And also SVMs and regressors are not included in these steps.

For Fast R-CNN, the image is only passed through CNN once to get a feature map, and a pooling layer extracts the features, and then it's passed through the fully connected layer to get the output. And Faster R-CNN is even faster, because it adds a region proposal network that has the same convolutional layers, which can directly output regions proposal from the feature map. It is the fastest because the proposals are generated way faster.

2.

So, semantic segmentation requires us to look at each pixel closely so we know what it is and where it is. If we only use fully connected layers, which can blur the spatial information, but fully convolutional layers can retain that spatial information.

Another reason is that fully convolutional layers can take inputs of any size, there is no longer a fixed size requirement from the fully connected layers, this is more efficient to generate what we desire and is more applicable to real world settings.